

Nathan Perez

CSCI 4345-03

4/28/2025

Report

1. Explanation of project:

- Overview

This project is designed to use my networks Virtual Machine as a pseudo Network Attached Storage to store .mp3 audio files for music. The VM will then act as a server and host a media server known as Emby, which will import the audio files and be available to stream whenever the VM is connected. This is relevant to computer networks because it showcases many applications of networking such as server-client relationships, secure file transfers over a network, and streaming services, which are all interconnected over networks. Someone might want to build this network solution if they consume a lot of media, but though this project handles music, Emby allows for the implementation of other forms of media like movies and games for example. Even if someone wants to forgo implementing the Emby media server, having your own NAS in the VM is extremely useful for storing files securely and being able to access them from anywhere on any device, as long as you're connected to your VM.

- Motivations

This project is useful for anyone who wants a private and secure way to stream any media they may want. In this particular project I'm doing music but it can be used for multiple other forms of media. It's mainly useful for people who consume a lot of media and want to be able to stream it through their own server. However, even just the file transfer using Cyberduck is extremely useful on its own since it securely transfers files to your VM and can be used as a NAS that can also be accessible anywhere as long as you connect to your VM, it serves as your own drop-box.

2. Breakdown of your Implementation:

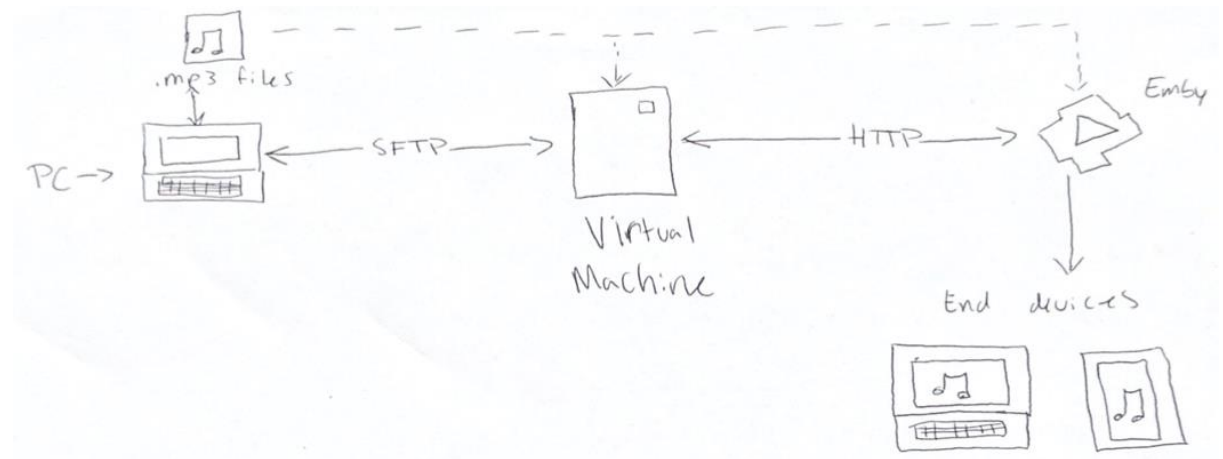
- Network Topology:

The layout of the network consists of one VM that is used as a server that can be used as a file storage as well, and host the Emby media server. Cyberduck is a client that handles a connection to the VM in order to use the SFTP feature and transfer files from your local computer to the storage in the VM.

- Configuration Details:

The first thing you need to do is create a VM on Azure and make sure you have an inbound rule on port 22 to allow SSH access to your VM. You need to create a directory for music that way you can store your audio files in there. You also need to configure the public IP address, port number, and login credentials of your VM on Cyberduck in order to use Cyberduck's SFTP. To configure Emby you also need to create an inbound port rule on port 8096 which is the port for Emby and allows for you to access the application.

- Communication Flow:



- Tools and Technologies:

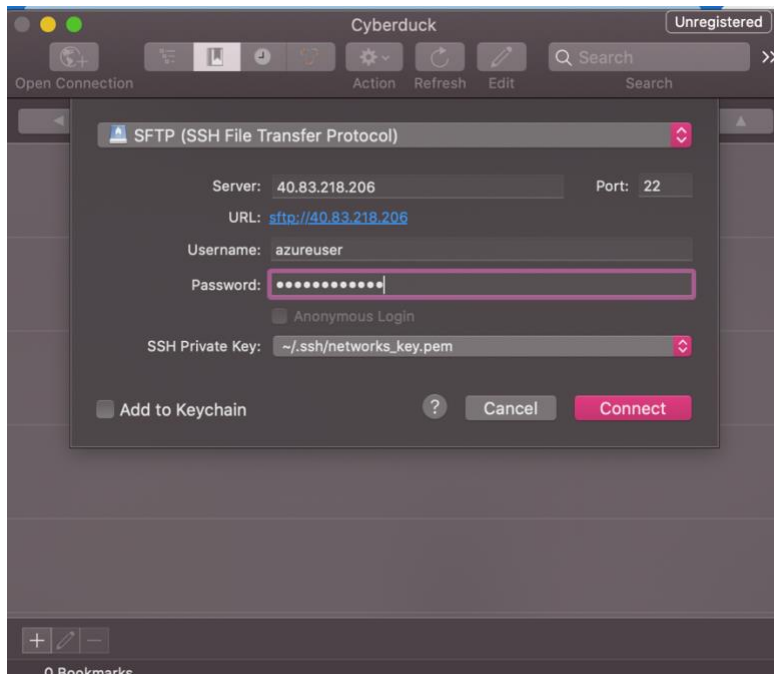
Azure was used to set up the virtual machine, Cyberduck was used to transfer the audio files to the VM music directory, and Emby was used to upload the .mp3 files to stream.

3. Demonstration

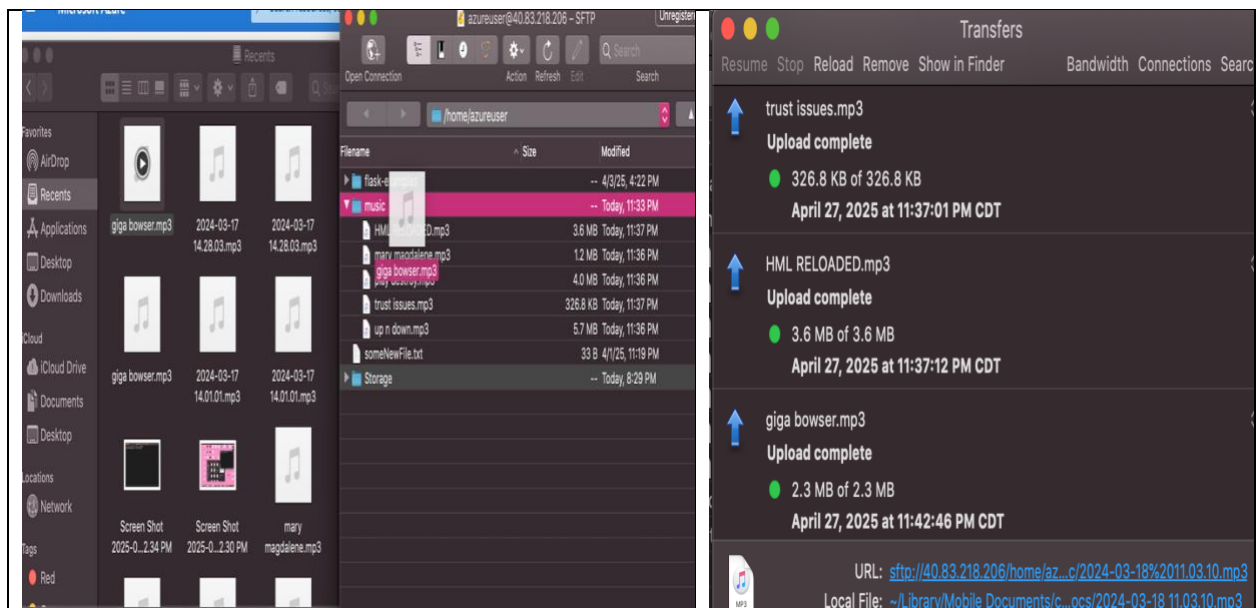
- Launching virtual machine

```
nathanperez — zsh — 80x24
Last login: Sat Apr 26 12:50:10 on ttys000
nathanperez@Nathans-Air ~ % ssh netvm
```

- Using Cyberduck's SFTP to transfer audio files from local computer to music directory (forgot to screenshot terminal when I created a music directory using `mkdir music`)



- **Dragging audio files from local computer to music directory in VM**



- **Checking music directory to make sure files transferred**

```
nathanperez — azureuser@networks: ~/music — ssh netvm — 80x24
azureuser@networks:~$ cd ~/music
azureuser@networks:~/music$ ls
'HML RELOADED.mp3'  'mary magdalene.mp3'  'trust issues.mp3'
'giga bowser.mp3'  'play destroy.mp3'    'up n down.mp3'
azureuser@networks:~/music$
```

- Downloading Emby media browser based on latest version

```
azureuser@networks:~$ wget https://github.com/MediaBrowser/Emby.Releases/releases/download/4.8.11.0/emby-server-deb_4.8.11.0_amd64.deb
--2025-04-29 02:14:42-- https://github.com/MediaBrowser/Emby.Releases/releases/download/4.8.11.0/emby-server-deb_4.8.11.0_amd64.deb
Resolving github.com (github.com)... 140.82.116.4
Connecting to github.com (github.com)|140.82.116.4|:443... connected.
HTTP request sent, awaiting response... 302 Found
Location: https://objects.githubusercontent.com/github-production-release-asset-2e65be/25282228/316b78a0-57a6-4077-8041-24ab36d8af11?X-Amz-Algorithm=AWS4-HMAC-SHA256&X-Amz-Credential=awsuser%2F20250429%2Fus-east-1%2Fs3%2Faws4_request&X-Amz-Date=20250429T021443Z&X-Amz-Expires=300&X-Amz-Signature=065551954dcc09746efdda526d7e90d5b66ccc6b5f6b0dca&X-Amz-SignedHeaders=host&response-content-disposition=attachment%3B%20filename%3Demby-server-deb_4.8.11.0_amd64.deb&response-content-type=application%2Foctet-stream [following]
--2025-04-29 02:14:43-- https://objects.githubusercontent.com/github-production-release-asset-2e65be/25282228/316b78a0-57a6-4077-8041-24ab36d8af11?X-Amz-Algorithm=AWS4-HMAC-SHA256&X-Amz-Credential=awsuser%2F20250429%2Fus-east-1%2Fs3%2Faws4_request&X-Amz-Date=20250429T021443Z&X-Amz-Expires=300&X-Amz-Signature=065551954dcc09746efdda526d7e90d5b66ccc6b5f6b0dca&X-Amz-SignedHeaders=host&response-content-disposition=attachment%3B%20filename%3Demby-server-deb_4.8.11.0_amd64.deb&response-content-type=application%2Foctet-stream
Resolving objects.githubusercontent.com (objects.githubusercontent.com)... 185.199.110.133, 185.199.111.133, 185.199.108.133, ...
Connecting to objects.githubusercontent.com (objects.githubusercontent.com)|185.199.110.133|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 173043244 (165M) [application/octet-stream]
Saving to: 'emby-server-deb_4.8.11.0_amd64.deb'

emby-server-deb_4.8 100%[=====>] 165.03M  35.2MB/s   in 4.5s

2025-04-29 02:14:48 (36.5 MB/s) - 'emby-server-deb_4.8.11.0_amd64.deb' saved [173043244/173043244]

azureuser@networks:~$ sudo dpkg -i emby-server-deb_4.8.11.0_amd64.deb
Selecting previously unselected package emby-server.
(Reading database ... 106677 files and directories currently installed.)
Preparing to unpack emby-server-deb_4.8.11.0_amd64.deb ...
Unpacking emby-server (4.8.11.0) ...
Setting up emby-server (4.8.11.0) ...
Created symlink /etc/systemd/system/multi-user.target.wants/emby-server.service -> /usr/lib/systemd/system/emby-server.service.
Processing triggers for libc-bin (2.39-0ubuntu8.4) ...
```

- Add inbound rule so that VM can run application on port 8096

 **Add inbound security rule** networks-nsg ×

Source ⓘ

Any

Source port ranges * ⓘ

*

Destination ⓘ

Any

Service ⓘ

Custom

Destination port ranges * ⓘ

8096 ✓

Protocol

☐ Any

☒ TCP

☐ UDP

☐ ICMPv4

Action

☒ Allow

☐ Deny

Priority * ⓘ

Add

Cancel

 Give feedback

- **Opening browser using `http://(VM public IP):8096`**

← → ↻

Not Secure http://40.83.218.206:8096/web/index.html?start=wizard#/wizard/wizardstart.html

☆ 🔍

< 

Welcome to Emby

This wizard will help guide you through the setup process of setting up Emby Server.

[Quick Start Guide](#)

Preferred display language

▼

Next →

- **Allow permissions for Emby to access music directory**

```

azureuser@networks:~$ sudo chown -R emby:emby /home/azureuser/music
azureuser@networks:~$ sudo chmod -R 755 /home/azureuser/music
azureuser@networks:~$ ls -l /home/azureuser/music
total 16740
-rwxr-xr-x 1 emby emby 3573483 Apr 28 04:37 'HML RELOADED.mp3'
-rwxr-xr-x 1 emby emby 2337835 Apr 28 04:42 'giga browser.mp3'
-rwxr-xr-x 1 emby emby 1180449 Apr 28 04:36 'mary magdalene.mp3'
-rwxr-xr-x 1 emby emby 3984240 Apr 28 04:36 'play destroy.mp3'
-rwxr-xr-x 1 emby emby 326776 Apr 28 04:37 'trust issues.mp3'
-rwxr-xr-x 1 emby emby 5727314 Apr 28 04:36 'up n down.mp3'
azureuser@networks:~$ sudo systemctl restart emby-server

```

- Bind music directory with Emby path so that the application can recognize the file

```

azureuser@networks:~$ sudo mkdir -p /var/lib/emby/music
azureuser@networks:~$ sudo mount --bind /home/azureuser/music /var/lib/emby/music
azureuser@networks:~$ sudo systemctl restart emby-server
azureuser@networks:~$

```

- Add a library folder /var/lib/emby/music

Folders

+ Add

/var/lib/emby/music



Music folder structure

Other or unstructured



This provides a hint to the library scanner about the way your folders are organized. If you're unsure, or if your folders don't exactly match one of the options, then please select the default option of Other.

Library Settings

Preferred metadata download language



Certification country

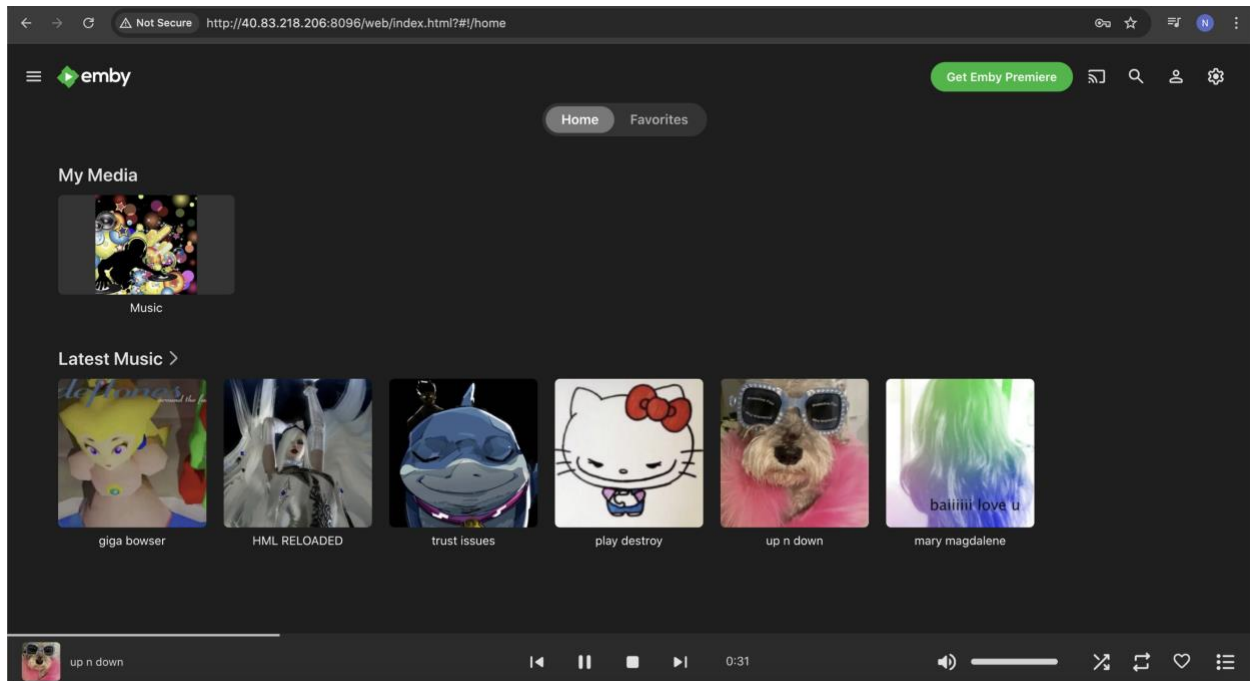


This will influence which content rating system is used.

Preferred image download language



- You now have access to your .mp3 files and can customize and add photos as you see fit



- Congrats! You can now stream your audio files on your own media server!

4. Citations

NetworkChuck, “How to Build a Raspberry Pi NAS (It’s AWESOME!!).” *YouTube*, 2

October 2021, www.youtube.com/watch?v=gyMpl8csWis. Accessed 26 Apr. 2025.

“Cyberduck | Libre Server and Cloud Storage Browser for Mac and Windows with Support for FTP, SFTP, WebDAV, Amazon S3, OpenStack Swift, Backblaze B2, Microsoft Azure & OneDrive, Google Drive and Dropbox.” *Cyberduck.io*, <https://cyberduck.io>.

Hosted~FTP~ Resources. “How to Use Cyberduck Tutorial - Connecting with FTP, FTPS, SFTP, Uploading and Downloading.” *YouTube*, 8 May 2017, www.youtube.com/watch?v=UYDWOvyzoAQ. Accessed 27 Apr. 2025.

MediaBrowser. “Releases · MediaBrowser/Emby.Releases.” *GitHub*, 20 Apr. 2025,
<https://github.com/MediaBrowser/Emby.Releases/releases>. Accessed 28 Apr. 2025.

SidneyUA. “Emby Server Can’t See Folders.” *Emby Community*, 22 Oct. 2020,
<https://emby.media/community/index.php?/topic/91173-emby-server-cant-see-folders/>.
Accessed 28 Apr. 2025.

OpenAI. *ChatGPT*. 28 Apr. 2025, <https://chat.openai.com/>. Accessed 28 Apr. 2025.